Nothing Sees the Abyss but Sound: A Physical Sensing Budget and an Open-Set AI for Discovering Unknown Species Below 4,000 m

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Abstract

Could satellites, or any remote sensor, reveal unknown animals in the abyssal and hadal ocean, and could an AI recognise a creature nobody has catalogued? We answer the first question with closed-form physics and the second with an open-set learning study. Seawater ends every remote modality within a few hundred metres: ocean-colour imagery integrates the top 45 m, an ideal noise-free two-way optical link would reach 576 m in the clearest water, the skin depth of a 10 GHz radar is 2.5 mm and even 3 Hz submarine radio is attenuated to nothing by 1.0 km. Only sound crosses the abyss: a 180 dB baleen call at 50 Hz is detectable at 121 km under deep-water spreading, a 210 dB odontocete click at 12 km, a 130 dB fish chorus at 115 m, while environmental DNA carries a signal 396 m downstream with a 20 h memory and a bioluminescent flash is visible 36 m from an in-situ camera. The eyes must be in the water; the satellite's role is to say where to put them. We then simulate a deep-sea soundscape of parametric call families (tonal, click, drum, burst) with species-specific parameters, Thorp absorption and Wenz noise, train a small CNN on 40 known species and ask six novelty scores whether a held-out clip belongs to one of 15 unknown species. Embedding-space scores (Mahalanobis, Mahalanobis) reach AUROC 0.86 (95% CI 0.85–0.87) where max-softmax gives 0.73 and an autoencoder 0.73; novelty degrades gracefully with range (SNR 36 to 20 dB: Mahalanobis 0.89 to 0.75). Clustering the flagged clips recovers the true grouping of two or three new species (ARI 0.91, 0.52) but merges larger sets. On real marine-mammal calls (Watkins, 31 species, 8 held out, 1668 clips) a CNN trained from scratch reaches only 0.66 AUROC (max-softmax); swapping in a frozen pretrained audio embedding (AST) in place of the from-scratch CNN, with no fine-tuning, raises it to 0.84 (energy), a gain positive in every one of 10 held-out draws (paired $\Delta = +0.125$) – confirming that real bioacoustics were embedding-limited, not scoring-limited. Finally, Chao1 tracks the unseen richness of a 1000-species log-series community as sensor-hours grow, the Good–Turing probability that the next detection is a new species falls from 12.2% at 10 h to 0.2% at 1000 h, and a greedy allocation of a fixed sensor budget over a satellite-derived habitat prior finds 12% more species than a uniform survey when the prior is 90% right and none when it is a coin flip.

1 Introduction

The deep ocean is the largest habitat on Earth and the least catalogued. Below the abyssal boundary at 4,000 m, and in the hadal trenches that reach 11,000 m, most expeditions still return with species new to science [12]. The question that motivates this paper is a natural one for the age of satellites and machine learning: *could we find those species from above — from orbit, or at least from a distance — and could an AI recognise an animal that no one has ever described?*

The first half of the question is a physics question, and we answer it before building anything. Seawater is opaque to almost everything a satellite can emit or receive. Sunlight is gone within a few hundred metres, radar and radio are absorbed within centimetres to metres, and gravity or magnetic anomalies respond to rock, not to animals. What a satellite *can* see is the surface: temperature fronts, chlorophyll, and — through the sea-surface height that gravity imposes — the bathymetry of seamounts and trenches where life concentrates. Section 3 quantifies each modality with closed forms: Beer–Lambert attenuation for light, the skin depth for electromagnetic waves, the passive sonar equation with Thorp absorption and Wenz ambient noise for sound, and a shed–decay–transport plume for environmental DNA. One modality survives the trip: low-frequency sound. The eyes of a deep-sea discovery system must therefore be hydrophones, samplers and cameras in the water, and the satellite's job is to say where to put them.

The second half is a learning question with an unusual twist. A classifier trained on known species will confidently assign a new species to the closest old one; discovery requires the model to say "none of the above" — the *open-set* recognition problem [7, 10, 11, 15]. Since no public recording of undescribed deep-sea animals exists (by definition), we build a simulator of a deep-sea soundscape: species are parametric call families with

species-specific parameter distributions, propagated to a hydrophone with the physics of Section 3. We train on K known species, hold out U unknown ones, and ask six novelty scores to separate them; we then cluster the clips flagged as novel and ask whether the clusters correspond to the unknown species — the step from "something new" to "how many new things". To keep the simulator honest we run the same pipeline on real marine-mammal recordings from the Watkins database in the BEANS packaging [5].

The last section closes the loop with survey statistics: given detections, how many species remain unseen (Chao1, Good–Turing [2, 4]), and how should a fixed budget of sensor-hours be spread over sites when a satellite-derived habitat prior is only sometimes right? The paper's contribution is thus a quantified map — what can be sensed, what can be recognised as new, and where to look — rather than a claim of discovery; every number regenerates from one command.

2 Related work

Ocean optics and remote sensing. Diffuse attenuation coefficients of the Jerlov water types [9, 14] set the penetration of sunlight; ocean-colour sensors integrate roughly one attenuation length. Satellite altimetry recovers bathymetry from the gravity signature of seafloor topography [13], which is how most seamounts are known — a habitat prior, not an animal detector.

Underwater acoustics. The passive sonar equation, Thorp's absorption formula and the Wenz ambient-noise curves are textbook material [17, 16, 18]. Low-frequency baleen calls are routinely detected at hundreds of kilometres on ocean-bottom hydrophones; deep-sea soundscapes are sparse and poorly characterised.

Environmental DNA. eDNA decays over hours to days and is transported by currents, so a sample integrates a footprint of hundreds of metres to kilometres [6]; it detects presence, not behaviour, and only for taxa with reference sequences — a limitation for discovery that we return to.

Bioacoustic machine learning. The BEANS benchmark [5] packages the Watkins Marine Mammal Sound Database with standard splits; strong results there rely on pretrained audio backbones. Our from-scratch CNN is deliberately minimal to keep the local pilot reproducible on a laptop, and its weakness on real data is part of the result; the GPU-scale run adds a direct comparison against a frozen, off-the-shelf pretrained backbone on the identical clips and splits, rather than only citing that pretrained backbones help elsewhere.

Open-set recognition. Max-softmax [7], energy [11], Mahalanobis [10] and deep nearest-neighbour [15] scores are the standard families; reconstruction error from an autoencoder is the classifier-free alternative. Grouping the novel samples is open-world recognition [1]; we use agglomerative clustering with silhouette model selection and score it with the adjusted Rand index [8].

How many species remain. Fisher's log-series [3] describes communities with many rare species; Chao1 [2] and Good–Turing coverage [4] estimate unseen richness and the probability that the next sample is new. Allocating survey effort under uncertainty is a value-of-information problem; we use a greedy marginal-gain rule.

3 The physical sensing budget

Figure 1 summarises the closed forms; all parameters are in the configuration.

Light. Irradiance decays as $E(z) = E_0 e^{-K_d z}$. In the clearest Jerlov type I water ($K_d = 0.022\,\mathrm{m}^{-1}$) the 1% euphotic depth is 209 m, and in coastal type III it is 46 m. Ocean-colour imagery integrates about one attenuation length, 45 m. Even an ideal detector with a 10^{-11} dynamic range on a round trip (sunlight down, reflection up) would reach 576 m in type I and 127 m in type III — an order of magnitude short of the abyss. In situ, a 10^{10} photon bioluminescent flash is visible to a $10\,\mathrm{cm}^2$ camera at 36 m: cameras see what swims past, nothing more.

Radio and radar. The skin depth $\delta = 1/\sqrt{\pi f \mu_0 \sigma}$ with $\sigma = 4\,\mathrm{S/m}$ is 2.5 mm at 10 GHz, 25 cm at 1 MHz and 145 m at 3 Hz; with a generous 60 dB link margin the 3 Hz extremely-low-frequency band used to signal submarines reaches 1.0 km. No satellite electromagnetic sensor sees below the mixed layer.

Sound. The passive sonar equation $\mathrm{SL} - \mathrm{TL}(r, f) - (\mathrm{NL}(f) - \mathrm{DI}) \geq \mathrm{DT}$ with Thorp absorption (0.003 dB/km at 50 Hz, 4.1 dB/km at 20 kHz) and Wenz noise at sea state 3 gives the ranges of Table 1. A 180 dB baleen call at 50 Hz is detectable at 23 km with spherical spreading and 121 km when spreading becomes cylindrical beyond the water depth; a 210 dB odontocete click at 20 kHz is absorption-limited to 12 km; a 130 dB fish chorus at 300 Hz reaches 115 m and a 100 dB soft-bodied animal only 3 m. Sound is the one modality that crosses the abyss, and it does so for loud, low-frequency callers; the quietest animals are audible only from a few metres, which is where an instrumented lander or ROV must be.

Environmental DNA. A source shedding 10^8 copies/h into a 0.05 m/s current with eddy diffusivity

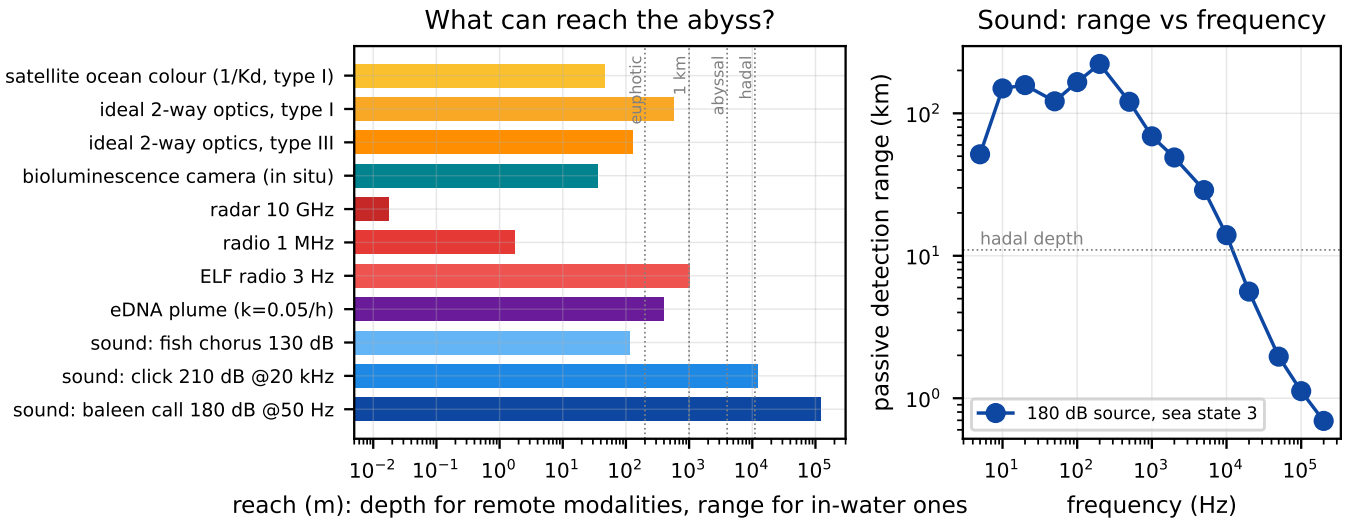


Figure 1: Left: how far each modality reaches (depth for remote sensing, range for in-water sensors), log scale, against the euphotic, 1 km, abyssal and hadal marks. Right: passive detection range of a 180 dB source versus frequency (sea state 3, hybrid spreading); only sound below a few hundred hertz crosses the full water column with margin.

source	SL (dB)	f	range
baleen call	180	50 Hz	121 km
odontocete click	210	20 kHz	12 km
invertebrate snap	150	5 kHz	1.2 km
fish chorus	130	300 Hz	115 m
soft body, quiet	100	1 kHz	3 m

Table 1: Passive detection ranges (sea state 3, moderate shipping, DT 10 dB, hybrid spreading with a 4 km transition).

$1\,\mathrm{m}^2/\mathrm{s}$ and decay $0.05\,\mathrm{h}^{-1}$ is detectable at 0.01 copies/L up to 396 m downstream, and the sample's memory is $1/k = 20$ h. eDNA thus integrates a neighbourhood of hundreds of metres over a day — a superb presence sensor, but one that identifies only what has a reference sequence, so for a truly new species it returns "unassigned" rather than a description.

What the satellite is for. Altimetry-derived bathymetry, surface chlorophyll and thermal fronts do not see animals, but they predict where animals concentrate (seamounts, canyons, upwelling). We carry this into Section 6 as a habitat prior of variable reliability: the satellite tells the ship where to lower the hydrophone.

4 Data and the open-set method

4.1 A synthetic deep-sea soundscape

A *species* is a call family with a species-specific parameter distribution, narrow enough that two species of the same family differ in a stable way: *tonal* (FM sweep with harmonics and vibrato; baleen moans, whistles), *click* (train with a characteristic inter-click interval and centre frequency), *drum* (pulsed swim-bladder knocks) and *burst* (band-limited noise pulses). Each clip draws a call realisation with ±12% parameter jitter, scales it to the species' source level (150–185 dB re 1 μPa), applies the transmission loss of Section 3 at a log-uniform range of 1–100 km, and adds Gaussian noise shaped by the Wenz spectrum at sea state 3. Features are 64-band log-mel spectrograms of 2 s at 16 kHz (Figure 2). The local study uses 40 known and 15 unknown species with 200 clips each (mean receiver SNR 26 dB).

4.2 Real calls

The Watkins Marine Mammal Sound Database "best of" cuts, in the BEANS packaging [5], provide 1668 clips of 31 species with at least eight clips each, after deduplicating by content hash across four of the database's five parquet shards (`train_low` is excluded: it is very likely a subset of the other train shard, and mixing it in risks the same clip appearing in both a training split and a held-out split). Clips are resampled to 16 kHz, cropped or padded to 2 s and featurised identically for the from-scratch CNN path; a small quantised cache is checked in so the pipeline reproduces offline. The pretrained-embedding path (Section 5) instead uses each clip's native duration up to 10 s, matching the backbone's own training length.

4.3 Models and novelty scores

A three-block CNN (16–32–64 channels, global pooling, 64-d embedding, softmax head) is trained with cross-entropy on the known species only; a convolutional autoencoder (32-d latent) is trained on the same clips; and a PCA(40) + multinomial logistic pipeline is the shallow baseline. On held-out known clips and all unknown clips we compute six novelty scores: 1−max softmax [7], energy $-\log\sum e^{z}$ [11], cosine distance to the k-th nearest train-

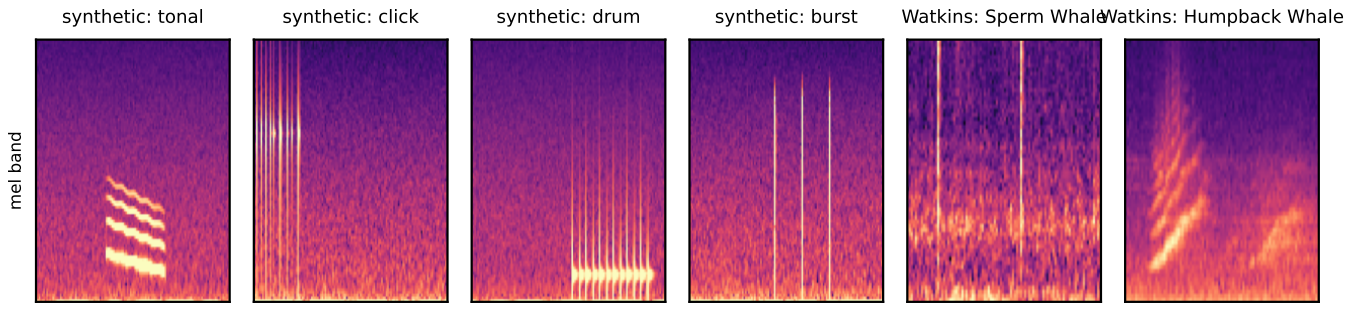


Figure 2: Log-mel clips (64 bands × 96 frames, 2 s). Four synthetic call families at 5 km range under Wenz noise, and two real Watkins clips.

ing embedding ($k = 5$) [15], minimum class-conditional Mahalanobis distance [10], autoencoder reconstruction error, and 1 − max probability of the PCA baseline. Metrics are AUROC and AUPR for unknown-vs-known and the false-positive rate at 95% recall of unknowns, plus the known-class accuracy.

4.4 From novel clips to candidate species

Clips whose score exceeds the 95th percentile of the known test scores are flagged; their embeddings are clustered by average-linkage agglomerative clustering on cosine distance, with the number of clusters chosen by the silhouette score over 2–10. We report the adjusted Rand index against the true unknown-species labels, the estimated versus true number of new species, and the precision and recall of the flagging step.

5 Can an AI tell it has never heard this before?

Synthetic soundscape. On the base condition (Figure 3, and the blue bars of Figure 4) the CNN reaches 0.92 closed-set accuracy, and the embedding-space scores separate unknown species from known ones with AUROC 0.86 (kNN, 95% bootstrap CI 0.85–0.87) and 0.86 (Mahalanobis), against 0.73 for max-softmax, 0.61 for energy, 0.73 for the autoencoder and 0.47 for the PCA baseline. The ordering is the one reported on image benchmarks: the geometry of the embedding knows more about novelty than the confidence of the head, and a reconstruction error trained only on known clips barely beats chance (best 0.80 across conditions) because a new call is still a call and reconstructs well. Even the best score has a false-positive rate of 46% at 95% recall of unknowns, which is why the flagging step below uses a high threshold and expects to miss some novel clips.

Range and data. Novelty degrades gracefully with distance: from a 10 km ceiling (SNR 36 dB) to 300 km (20 dB), Mahalanobis falls from 0.89 to 0.75 and kNN to 0.85, while max-softmax drops from 0.79 to 0.72. With only four known species the Mahalanobis score is still 0.95 (max-softmax 0.74), and with 15 training clips per species 0.76 (max-softmax 0.44): the distance-based scores are robust to small catalogues, the confidence-based ones are not, and for deep-sea work the catalogue *is* small.

Real calls, from-scratch CNN. The orange bars of Figure 4 are the honest part. A CNN trained from scratch on 1668 clips reaches only 0.70 closed-set accuracy on 31 species, and the best novelty score is max-softmax at AUROC 0.66; the embedding-space scores that dominate on synthetic data fall to 0.61 (kNN) and 0.61 (Mahalanobis), i.e. the embedding has not learned species geometry. Real recordings vary in recorder, distance, background and duration far more than our simulator.

Real calls, pretrained backbone. The green bars swap the from-scratch CNN for a frozen Audio Spectrogram Transformer embedding (`MIT/ast-finetuned-audioset-10-10-0.4593`), with no fine-tuning, feeding the same six scores through a deterministic sklearn probe instead of a second from-scratch classifier. Accuracy rises to 0.88 and the best score to 0.84 (energy), a paired improvement of $\Delta = +0.125$ that is positive in every one of the 10 holdout draws. This confirms the diagnosis directly: the embedding, not the scoring method, was the bottleneck on real data, and the fix is a frozen off-the-shelf model, not more deep-sea recordings (which do not exist at the scale a from-scratch embedding would need).

How many new species? Figure 5 takes the flagged clips (precision ≥ 0.66 at every U, recall 0.69–0.39) and asks whether they group into the unknown species. With two or three unknown species the clusters are nearly perfect (ARI 0.91 and 0.52), though the silhouette criterion over-splits to $\hat{k} = 11$; with five or eight unknown species the silhouette prefers two or three broad clusters (ARI 0.94, 0.76) — it groups the novel clips by call family rather than by species. This is the step where a human (or a second, finer model) enters: the system reliably says "here are the clips that are not in the catalogue, and they come in a few kinds"; counting the kinds precisely needs either better embeddings or the taxonomist. On

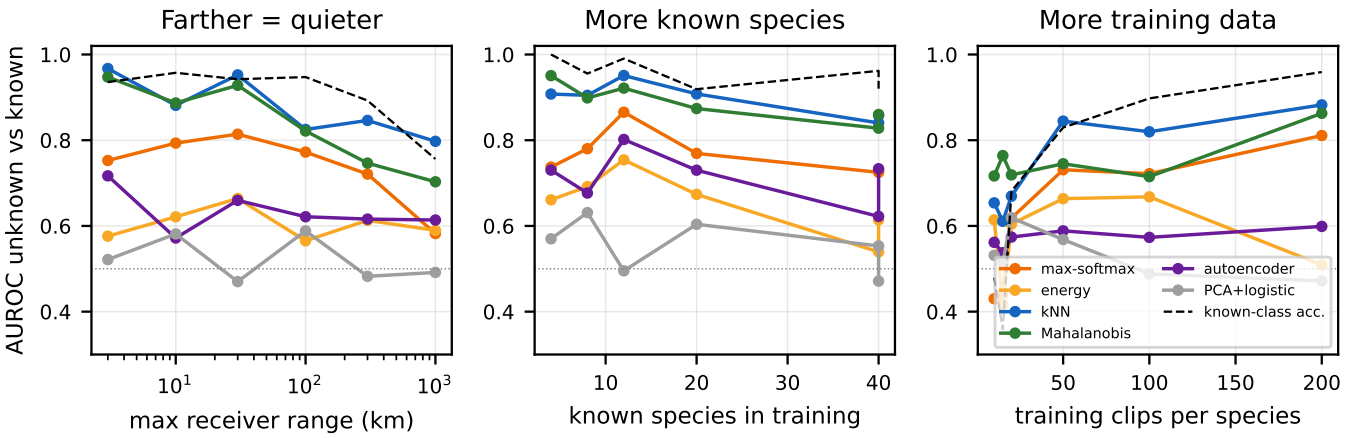


Figure 3: AUROC of six novelty scores (unknown vs known species) on the synthetic soundscape as the receiver range grows (left; farther means lower SNR), as the number of known species grows (middle) and as training clips per species grow (right). The dashed line is the closed-set accuracy on known species.

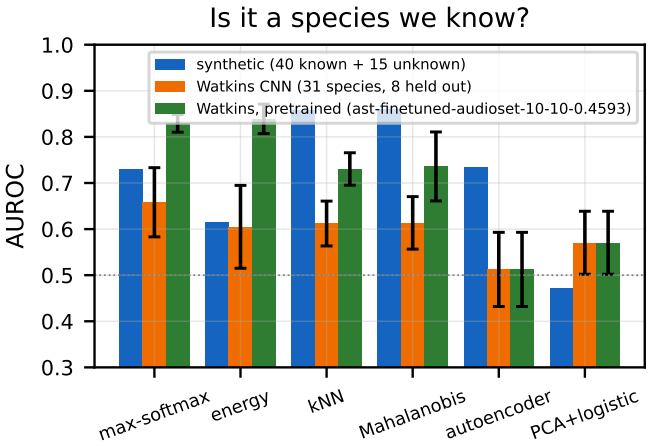


Figure 4: Novelty AUROC per score on the synthetic soundscape (base condition), on real Watkins calls with a from-scratch CNN, and on the same real clips with a frozen pretrained backbone (AST) in place of the CNN (31 species, 8 held out as unknown, mean ± s.d. over 10 holdout draws).

real data the same procedure gives a mean ARI of 0.33, consistent with the weak embedding.

6 How many remain, and where to look

How many remain. A community of 1000 species with log-series abundances is sampled by a sensor whose per-species detection rate scales with abundance. After 10 h it has seen 48 species and Chao1 estimates 94 ± 27; after 100 h, 121 seen and 172 ± 22; after 1000 h, 221 seen and 275 ± 21; after 3000 h, 272 seen and 326 ± 21 (Figure 6, left). Chao1 is a lower bound and behaves like one: it rises with effort and stays well below the truth while most species are still singletons, which is the regime the deep sea is in. The Good–Turing discovery probability — the chance that the *next* detection is a species not yet seen — falls from 12.2% at 10 h to 1.9% at 100 h and 0.2% at 1000 h. That curve is the honest answer to "can AI discover new species": the sensor finds new ones fast at first, and each further one costs exponentially more hours, whatever the classifier.

Where to look. Twenty sites belong to a seamount habitat (600 species) or an abyssal plain (120 species); a satellite-derived prior labels each site's habitat correctly with probability q. A greedy rule spends the 2000 h budget where the expected marginal number of new species under the prior is largest. When the prior is a coin flip ($q = 0.5$) greedy and uniform find the same 363 species; at $q = 0.7$ the gain is 10%; at $q = 0.9$ greedy finds 408 species against 364 (12%), and with a perfect prior 411 against 364 (13%). The satellite cannot see the animals, but a reliable map of where the rich habitat is buys a sixth more discoveries per sensor-hour — and an unreliable one buys nothing, which is a reason to spend on bathymetry before spending on hydrophones.

7 Limitations and conclusion

Limitations. The sensing budget uses textbook closed forms: spherical-then-cylindrical spreading rather than a ray or parabolic-equation propagation model, a single sea state and shipping level, a generic Wenz spectrum rather than a measured deep-sea soundscape, and an eDNA plume with illustrative shedding and decay rates. The synthetic species are four parametric families; real deep-sea sound includes many mechanisms we did not model and many animals that make no sound at all, for which acoustics offers nothing and only cameras, eDNA or trawls apply. The real-data result covers 1668 clips of 31 species over 10 holdout draws, comparing a from-scratch CNN against a frozen pretrained backbone; it demonstrates the gap to synthetic performance and how much of it a backbone swap closes, not the ceiling of the method – fine-tuning the backbone, or a bioacoustic-specific pre-

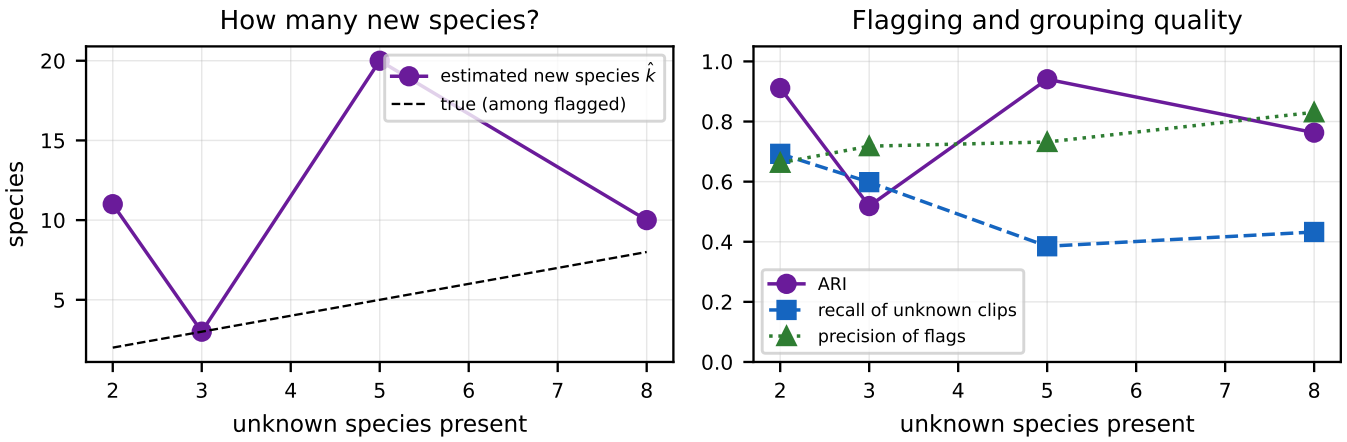


Figure 5: Clustering the flagged clips. Left: estimated number of new species (silhouette-selected $\hat{k}$) versus the number of unknown species actually present among the flagged clips. Right: adjusted Rand index of the clusters against the true unknown species, and the recall and precision of the flagging step.

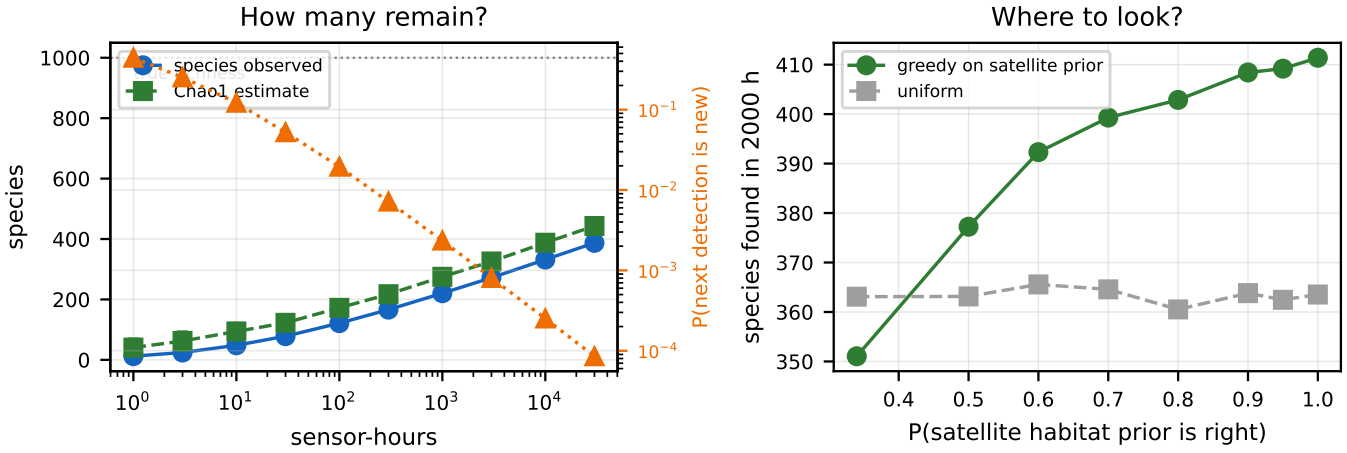


Figure 6: Left: species observed and the Chao1 estimate of total richness versus sensor-hours for a 1000-species log-series community (mean ± s.e. over 30 replicates), with the Good–Turing probability that the next detection is a new species (orange, right axis). Right: species found with a 2000 h budget over 20 sites by a greedy allocation on a satellite habitat prior versus a uniform survey, as the prior's reliability varies.

trained model, are natural further steps. Clustering uses one criterion (silhouette) whose failure at larger U we report rather than tune away. The survey model assumes detections are independent Poisson events with rates proportional to abundance, and the habitat prior's reliability is a free parameter. All of these are configuration knobs or data swaps.

Conclusion. Nothing sees the abyss but sound. Sunlight, radar and radio are gone within metres to a few hundred metres of the surface even for an ideal detector, so no satellite will ever image a hadal animal; the satellite's contribution is bathymetry and surface productivity, a prior for where to look that is worth 12% more species per sensor budget when it is reliable and nothing when it is not (and -3% — a loss — when it is actively misleading). Low-frequency sound crosses the full water column — a 180 dB call at 50 Hz is detectable at 121 km — while quiet animals must be within metres of an in-situ camera or an eDNA sampler. Given sensors in the water, an open-set model can tell that a call belongs to no catalogued species: distance-based scores on a learned embedding reach AUROC 0.86 on a physically simulated soundscape, degrade gracefully with range and catalogue size, and let the flagged clips be grouped into a handful of candidate species when there are few of them. On real marine-mammal calls, the same pipeline trained from scratch only reaches 0.66 — but swapping in a frozen, off-the-shelf pretrained audio embedding, with no fine-tuning and no more deep-sea data, raises it to 0.84, confirming that the bottleneck was the embedding, not the amount of real data or the scoring method. And whatever the model, the statistics of rare species set the pace — the probability that the next detection is new falls from 12.2% to 0.2% between 10 and 1000 sensor-hours. An AI can recognise the unknown; finding it is still a matter of where you drop the hydrophone, and for how long.

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